Study G — Appendix B: Trajectory Atlas

Every recovery the campaign found, fault by fault and state by state

17 August 2026

Contents

1 How to read these

Three views per regime, all drawn from the same set of solves:

- **Descent profile** — altitude against horizontal distance to the pad, one panel per fault, with all ten of that regime's initial conditions overlaid. The dashed line is the 30° glide-slope cone the descent has to stay above; trajectories that ride it are ones for which the approach geometry, not the fault, is the binding constraint.
- **Ground track** — the same paths seen from above, with the pad and the 15 m landing-gate circle.
- **One initial condition, sixteen vehicles** — a single shared state flown by every plant in the catalogue. This is the study's premise in one picture: same state, different vehicle, different future. The state drawn is the one the faults disagree about most.

Line colour is the outcome: **green** landed, **amber** flew but missed the touchdown gate, **red** × no trajectory found at all — drawn at the initial condition it started from, because a missing line and an unsampled state would otherwise look identical. Panel-title colour is the FTC fault structure (grey healthy, blue additive, orange multiplicative, green structural).

Trajectories include the engine-off ballistic settle from the contact altitude down to the surface, so each line ends where the vehicle actually touched down.

A note on what these are **not**: each line is a single open-loop optimal control solution computed with exact knowledge of the damaged plant. They show what trajectory *exists*, not what a controller flying with an estimated plant would achieve.

2 `approach` — On approach

High, descending, small dispersions — the vehicle is where a healthy descent would have put it, and the fault is the only thing wrong. Across all 16 plants this regime landed 140 of 160 cases.

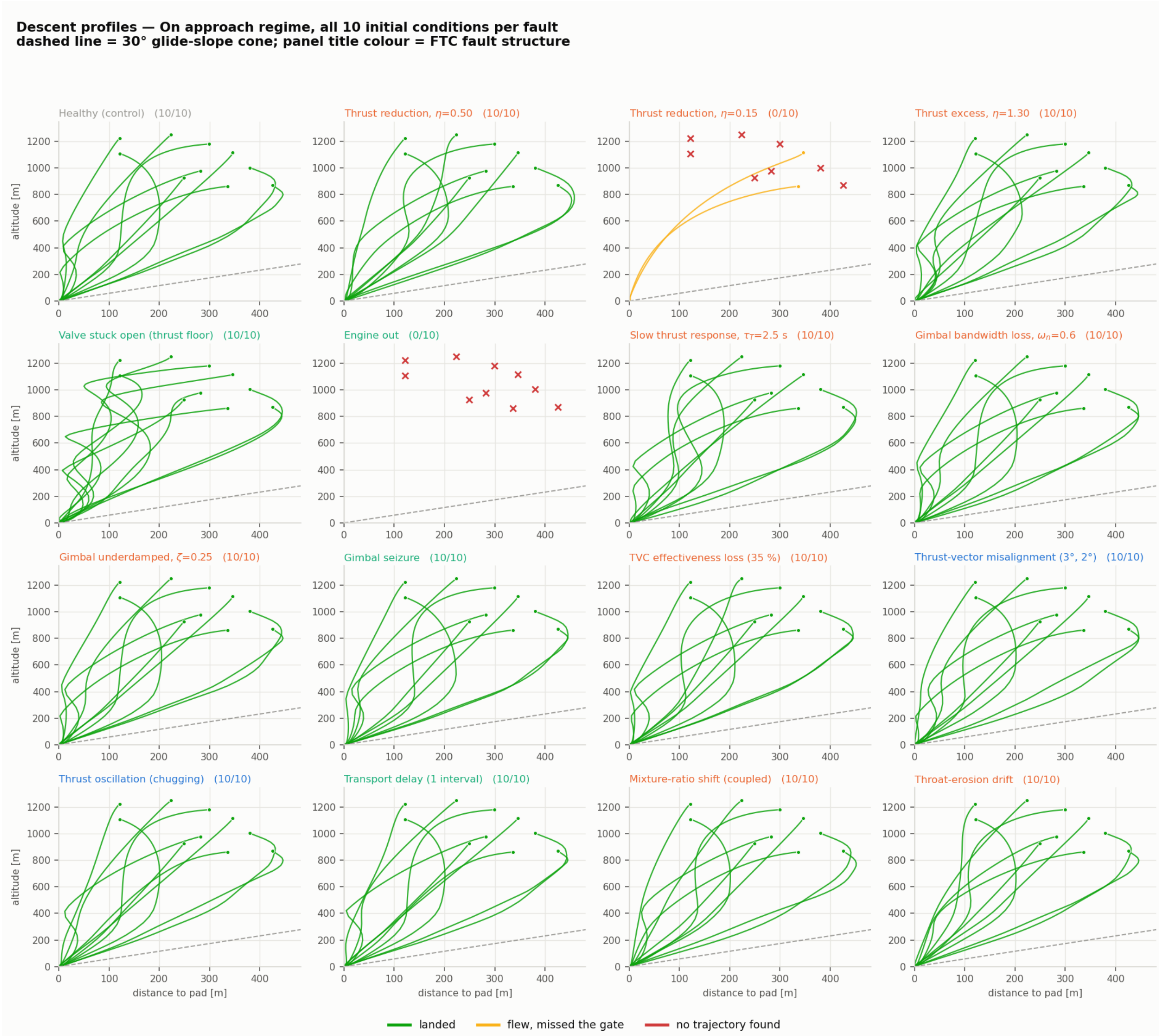

Figure 1: Descent profiles — On approach regime

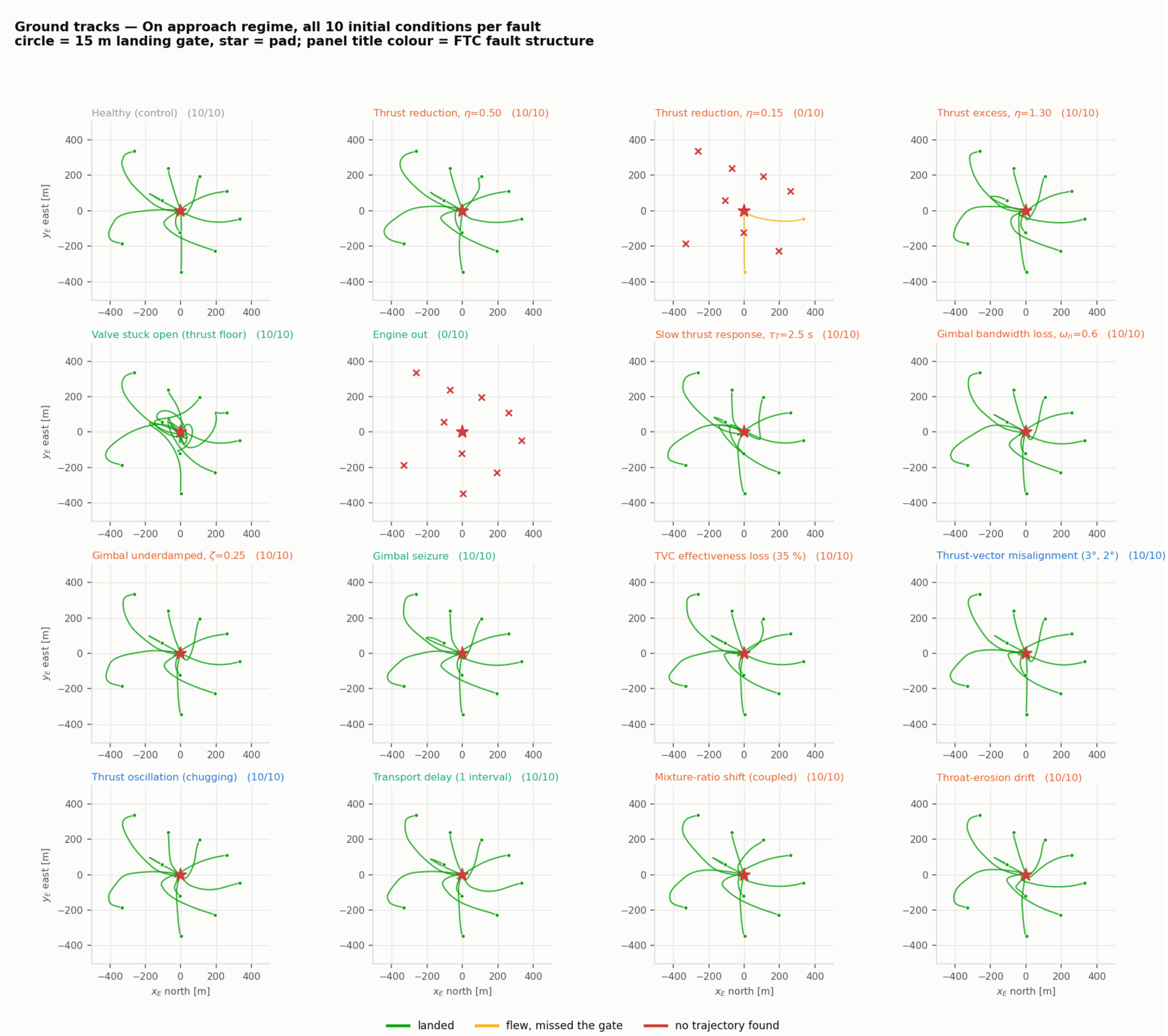

Figure 2: Ground tracks — On approach regime

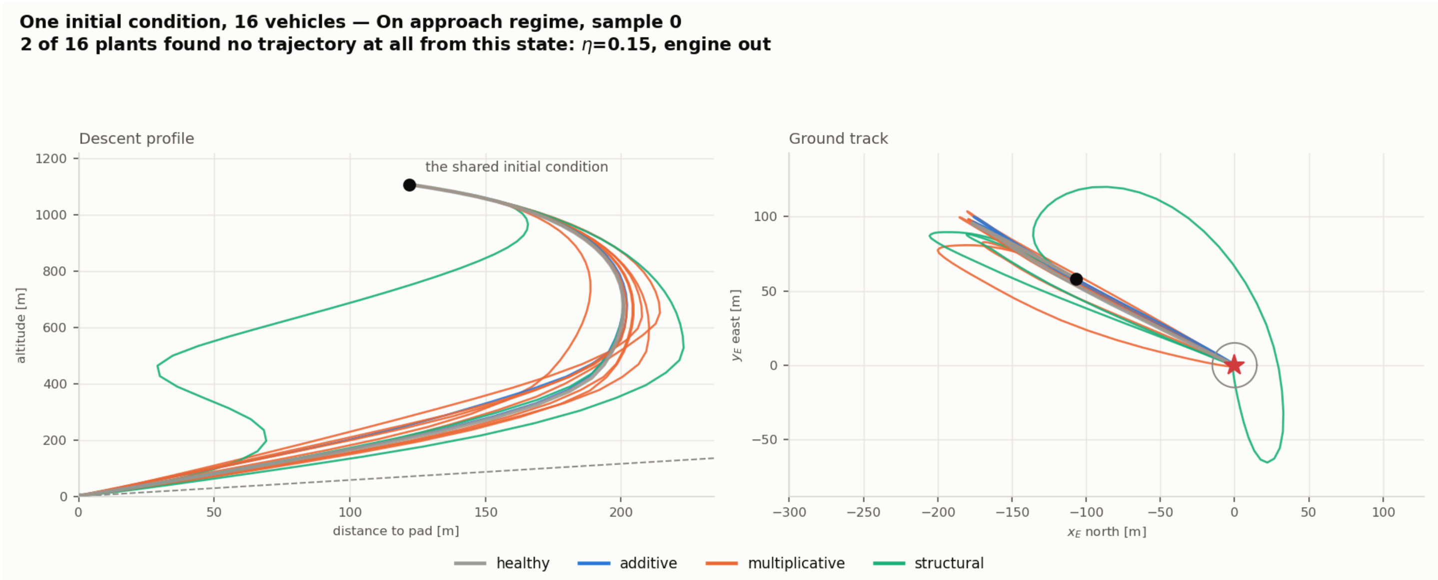

Figure 3: One initial condition, every plant — On approach regime

3 `dispersed` — Dispersed

Study f's wide box: position, velocity, attitude, rates and both sets of actuator states drawn over their full admissible ranges. Across all 16 plants this regime landed 124 of 160 cases.

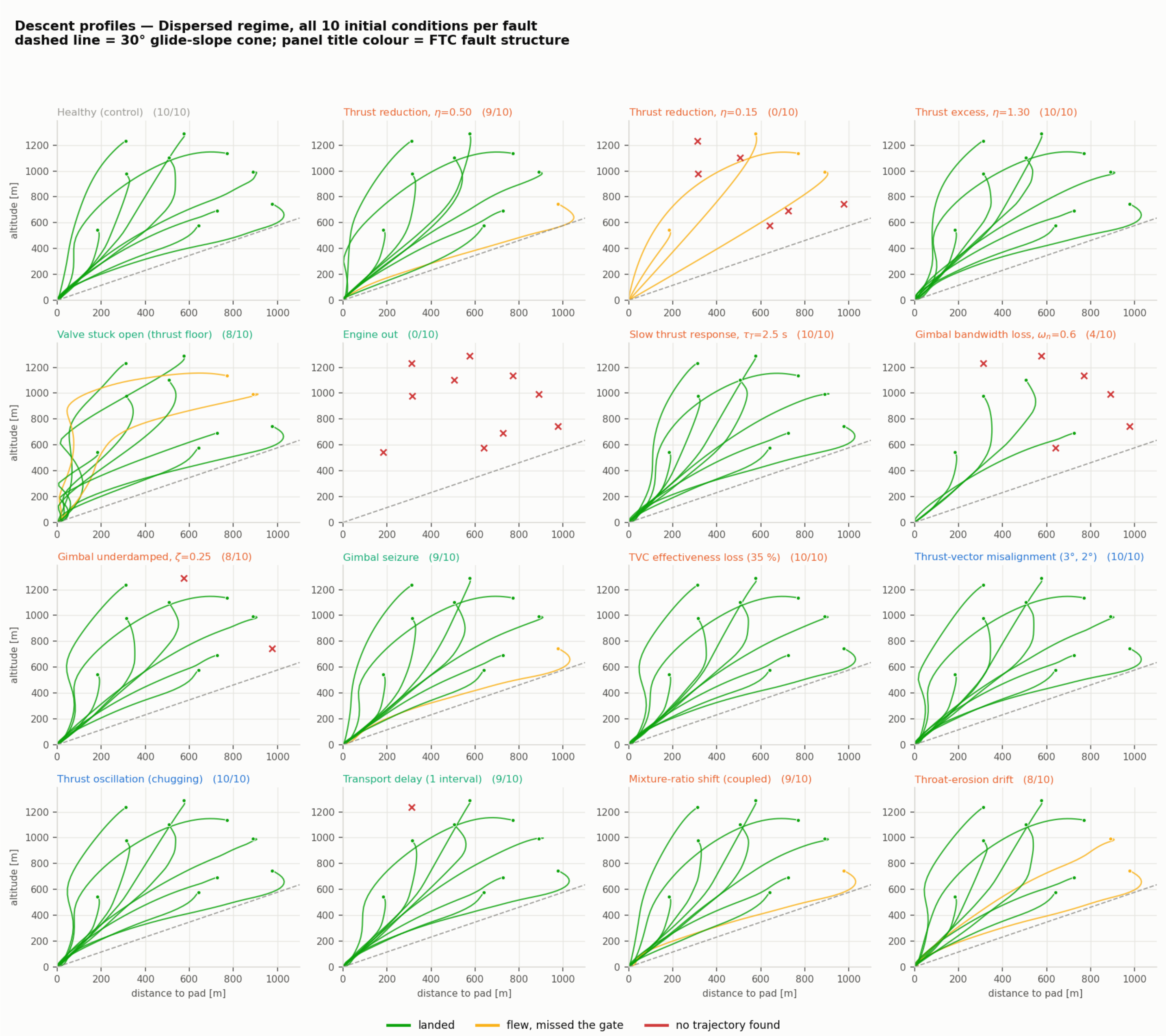

Figure 4: Descent profiles — Dispersed regime

Figure 5: Ground tracks — Dispersed regime

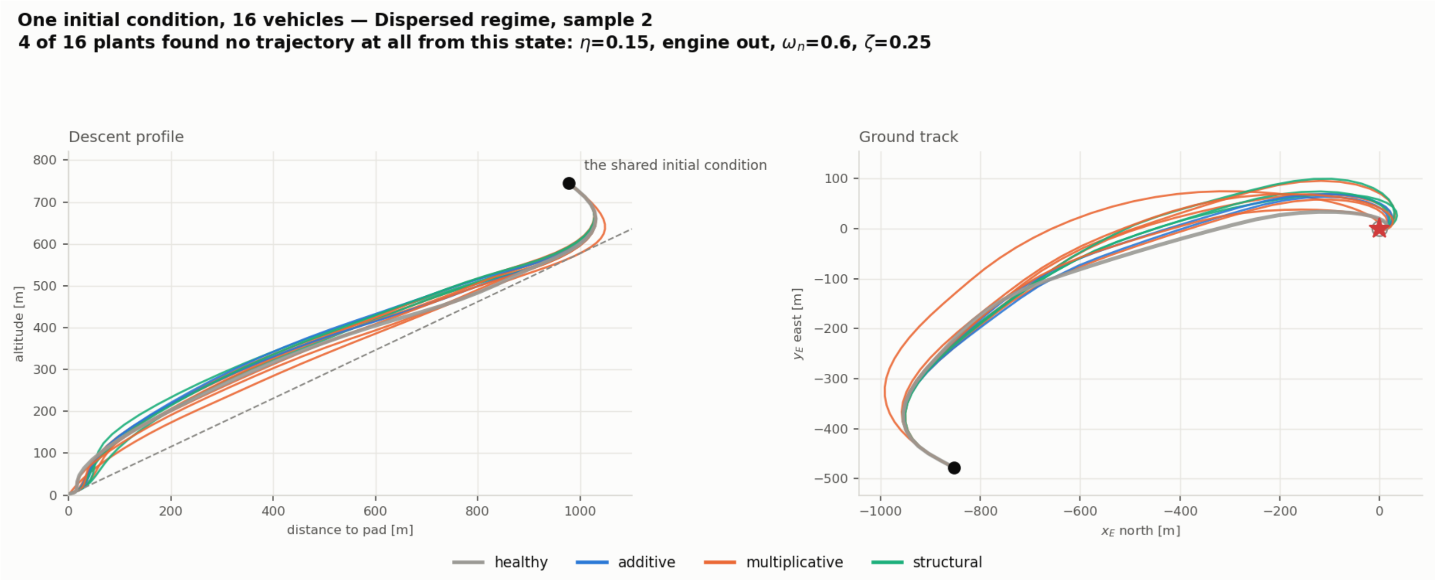

Figure 6: One initial condition, every plant — Dispersed regime

4 upset — Upset

The corner of the box where the vehicle is already rotating or tilted (>= 10 deg/s on some axis, or >= 15 deg of tilt) with the gimbals deflected — the states a fault transient actually produces. Across all 16 plants this regime landed 120 of 160 cases.

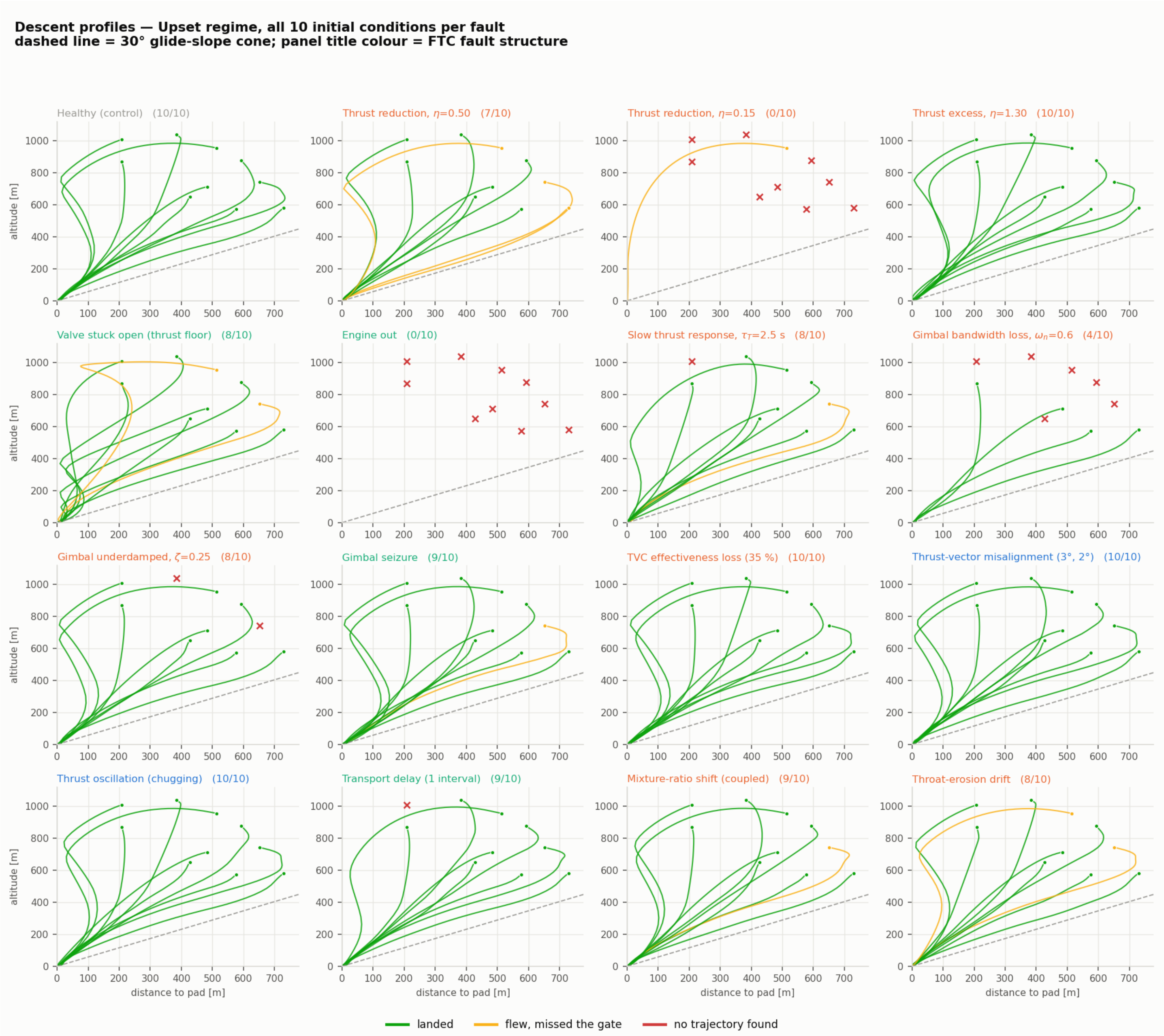

Figure 7: Descent profiles — Upset regime

Figure 8: Ground tracks — Upset regime

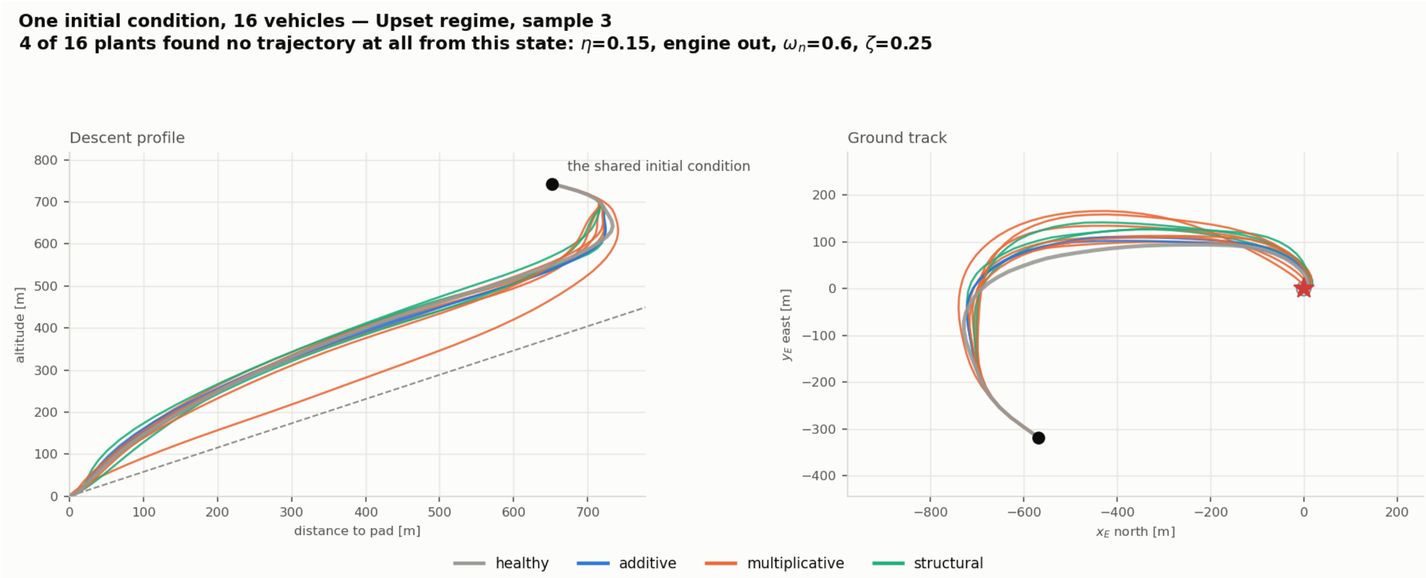

Figure 9: One initial condition, every plant — Upset regime

5 `low_late` — Low and late

Between 60 m and 200 m and still descending: the fault arrives when there is very little altitude left to trade for time. Across all 16 plants this regime landed 105 of 160 cases.

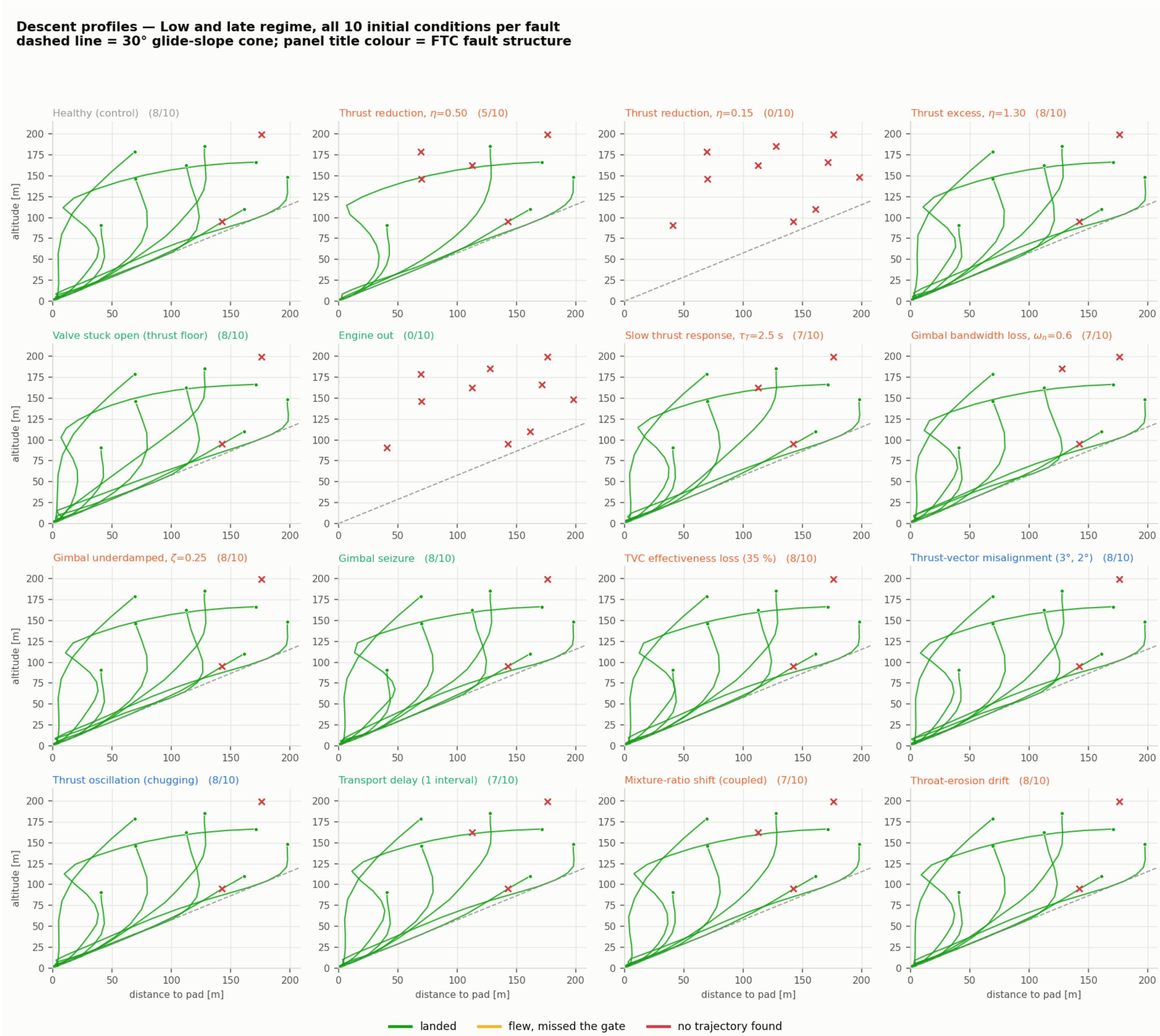

Figure 10: Descent profiles — Low and late regime

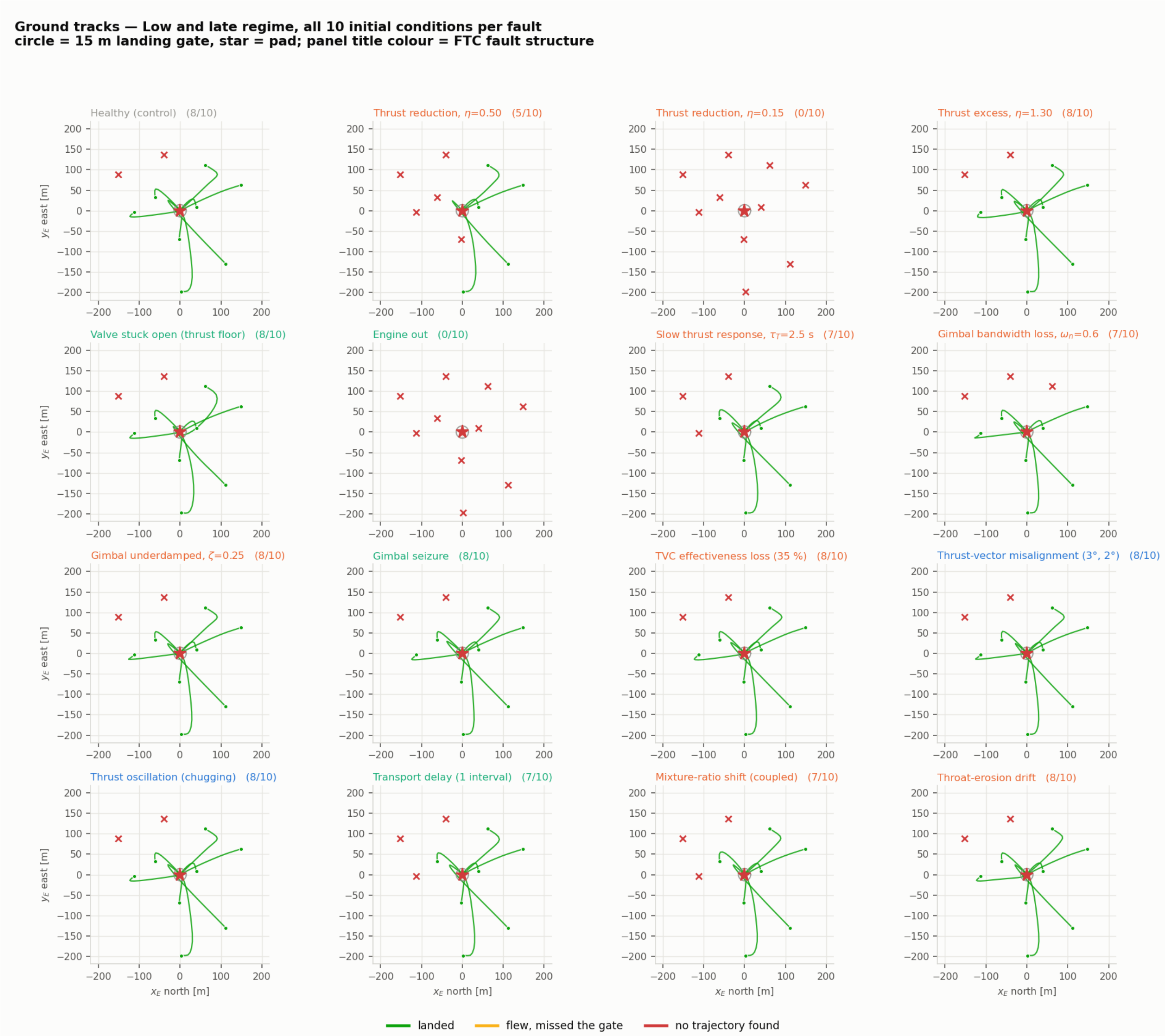

Figure 11: Ground tracks — Low and late regime

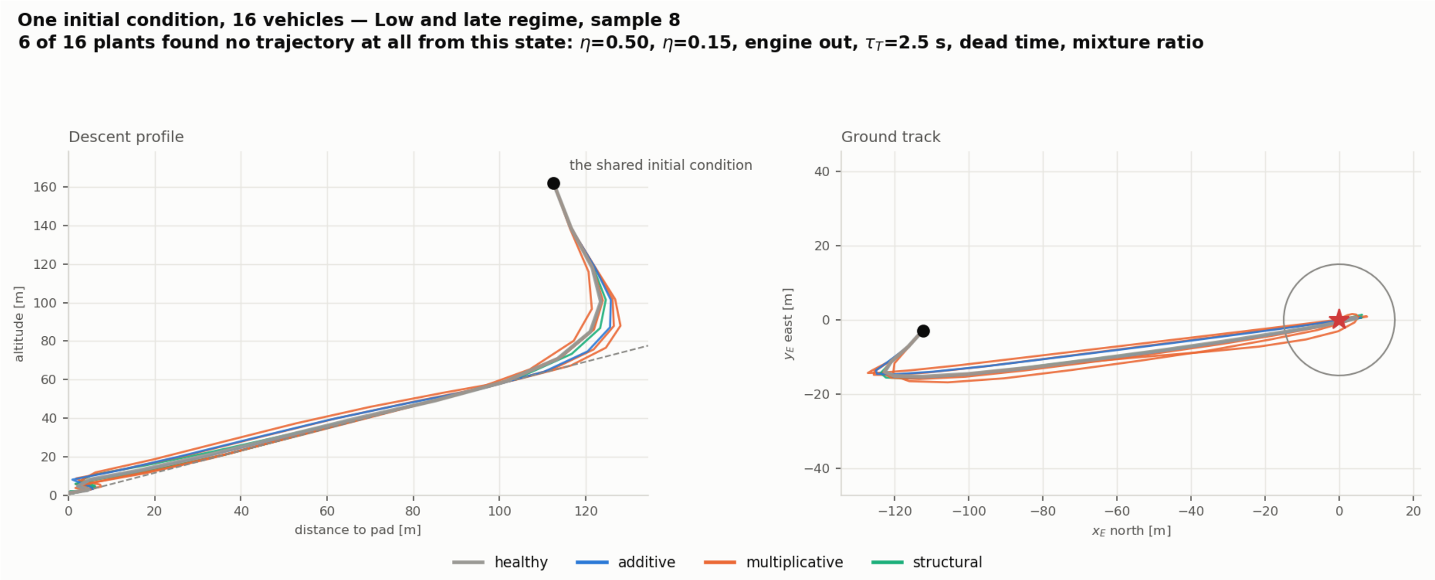

Figure 12: One initial condition, every plant — Low and late regime

6 `critical` — Critical

Below 140 m, descending hard, tilted and rotating — the regime where even the healthy vehicle frequently has no trajectory left, so the fault has to be paid for out of a margin that is already spent. Across all 16 plants this regime landed 66 of 160 cases.

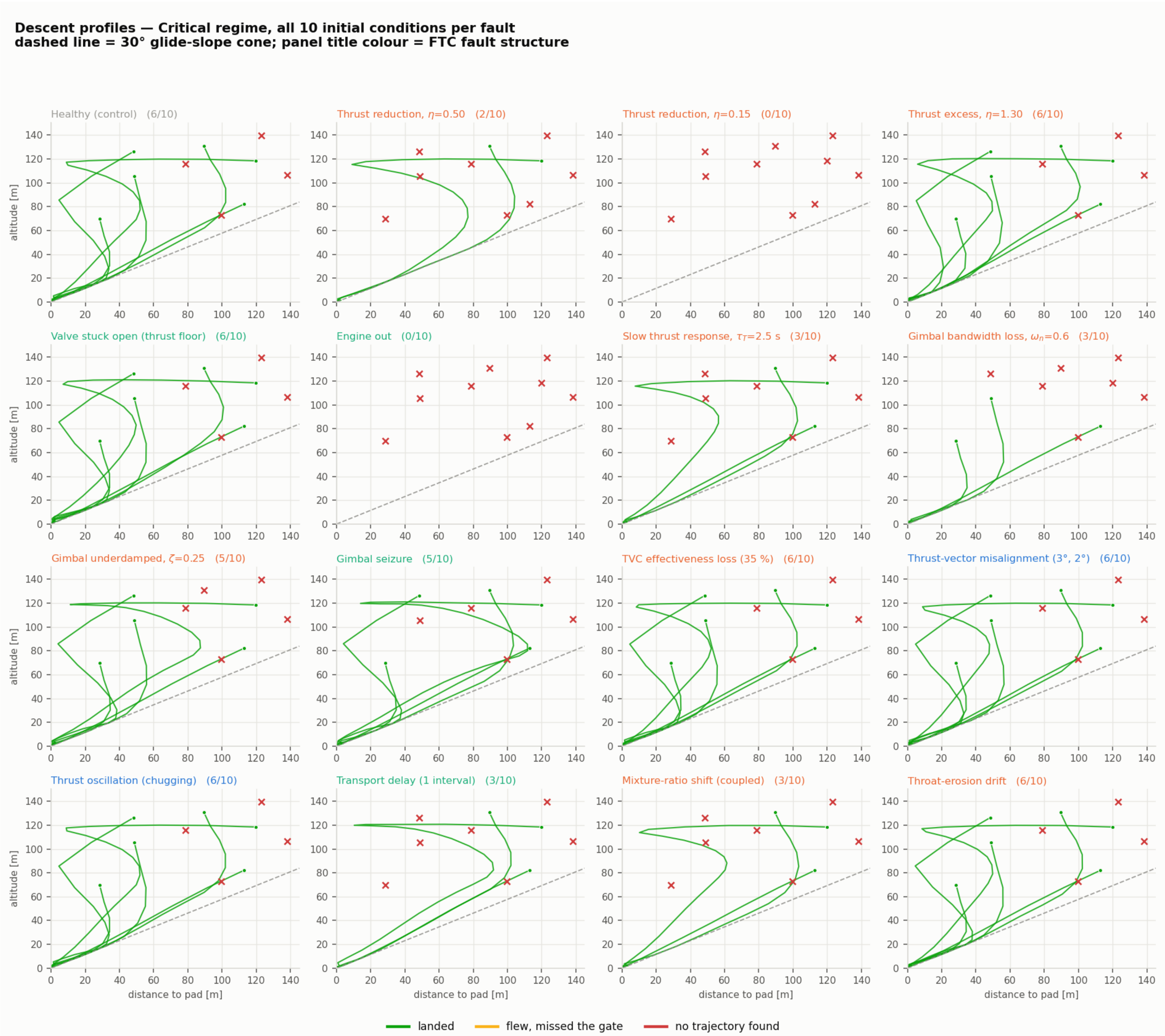

Figure 13: Descent profiles — Critical regime

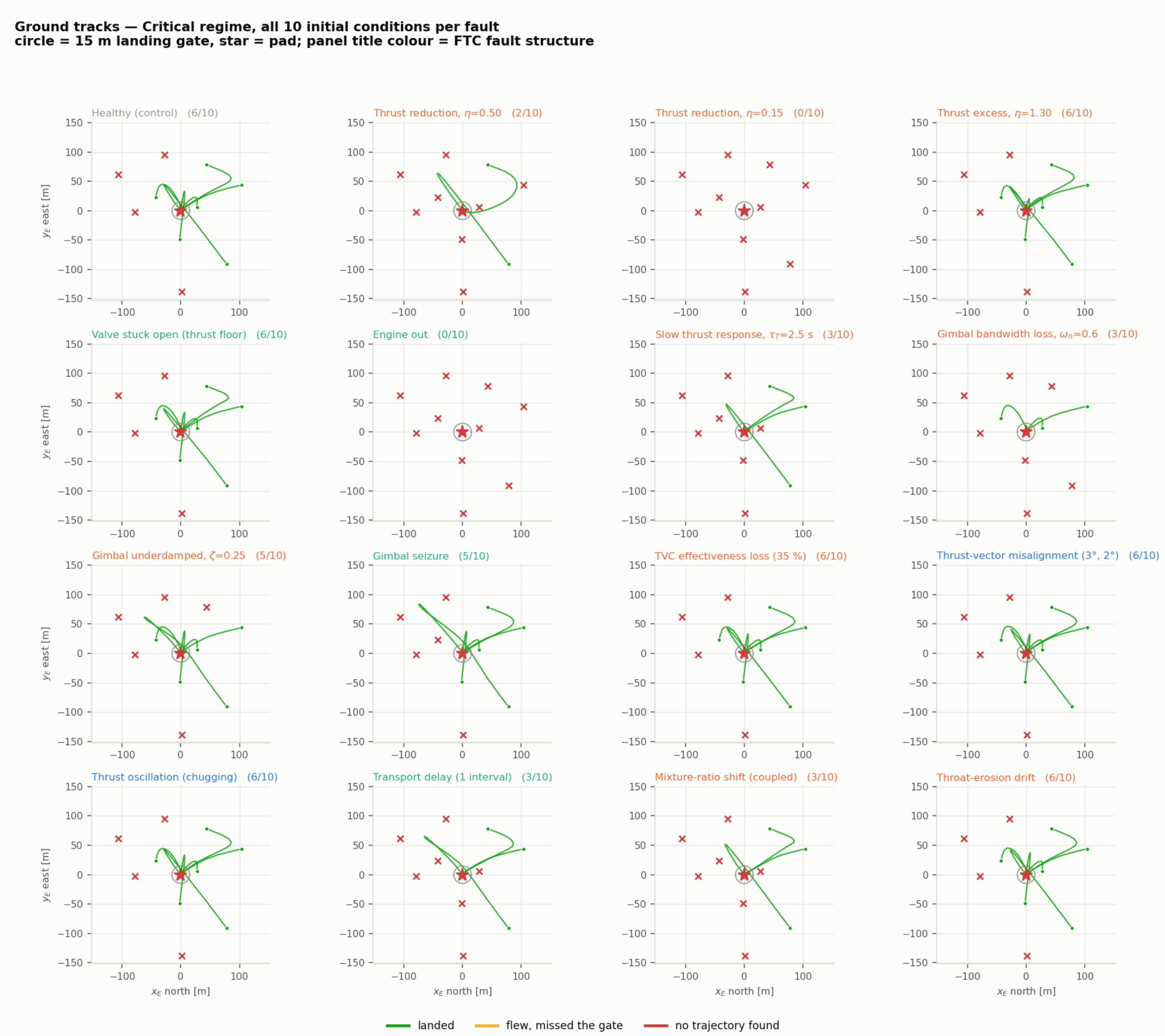

Figure 14: Ground tracks — Critical regime

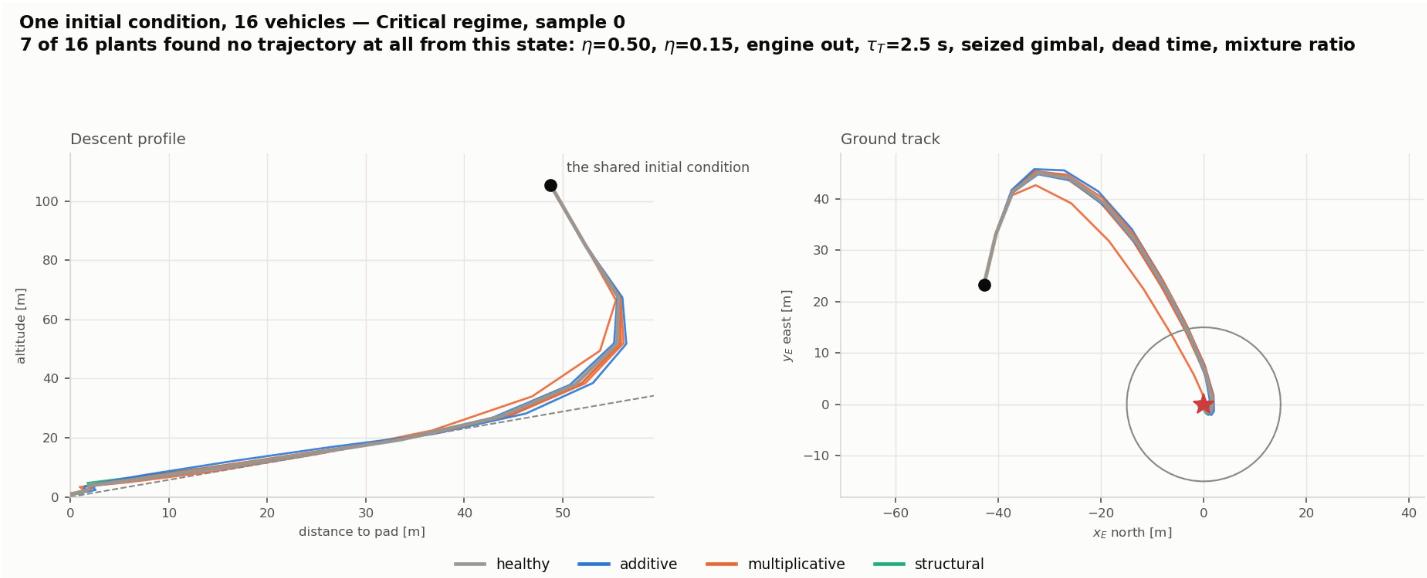

Figure 15: One initial condition, every plant — Critical regime